

M DHANUSH VARDHAN

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Summary

AI/ML Engineer Intern with hands-on experience in **machine learning**, **LLM integration**, and **AI-driven workflows**. Skilled in **Python**, **SQL**, **ML/DL techniques**, **LangChain**, **LangGraph**, and **MLOps practices**.

Experience

AI & ML Engineer Intern | WINIT SOFTWARE – On-site

Apr 2025 – Oct 2025

Automated Order Processing Escalation System

- Assisted in building end-to-end AI automation pipeline using **Gmail API**, **Mistral OCR**, and **Gemini LLM** to process **500+ daily purchase orders** from multiple document formats and email sources, achieving **95% extraction accuracy** in converting unstructured data into structured formats
- Supported development of **stateful orchestration workflows** using **LangChain** and **LangGraph** to manage extraction, validation, and escalation stages with decision-based branching, retry logic, and error handling capabilities
- Helped integrate **asynchronous processing** with **Celery** and **RabbitMQ** for non-blocking task execution, and implemented **PostgreSQL-based validation logic** with **21+ business rules** to detect pricing, UOM, and promotional discrepancies, contributing to **45% reduction in order errors**
- Contributed to building **role-based intelligent escalation system** with **Twilio** and **ElevenLabs AI-powered WhatsApp**, **email**, and **voice alerts** featuring interval-based retry logic, reducing **manual escalation effort by 60%**

Technologies: Python, Mistral OCR, Gemini LLM, LangChain, LangGraph, ElevenLabs AI, Twilio, PostgreSQL, Celery, RabbitMQ, Gmail API, Azure

AI & ML Engineer Intern | OPTIMANAGE SOFTWARE – Remote

Dec 2024 – Mar 2025

Virtual AI Mental Health Platform

- Developed **FastAPI-based backend pipeline** for multimodal AI psychologist with real-time speech processing, enabling **multilingual mental health support** with contextual memory and adaptive therapy generation
- Integrated **Azure Cognitive Services Speech-to-Text API** achieving **92% transcription accuracy** across **5+ languages** for AI-driven psychological reasoning and response generation
- Implemented **Azure OpenAI** and **LangChain workflows** with long-term memory using database storage to retrieve conversation history, analyze emotional patterns, and generate personalized therapy responses, improving **conversation relevance by 35%** and **user engagement by 50%**
- Built **phoneme-to-viseme mapping pipeline** for animated avatar lip-syncing with **98% accuracy** and **sub-200ms latency**, delivering natural multimodal conversational experience

Technologies: Python, FastAPI, Azure Cognitive Services, Azure OpenAI, LangChain, NLP, Speech Processing

Projects

FMCG Warehouse Risk Intelligence & Supply Chain Analytics | GitHub

Business Problem: FMCG companies faced weeks of manual analysis for warehouse risk assessment with data scattered across systems, causing delayed vulnerability identification and significant costs per incident.

Business Solution: Developed **GraphRAG system** transforming **25,000+ warehouse records** into knowledge graph, enabling natural language queries for automated risk assessment with multi-hop reasoning capabilities.

Achievements:

- Achieved **94.2% risk identification accuracy**, outperforming traditional methods by **22.7%**
- Reduced analysis time by **99.9%** with real-time risk assessment completing in **2.3 seconds**
- Demonstrated **2.4 crore potential savings** through proactive risk mitigation with **100% risk factor coverage**
- Built **end-to-end AI pipeline** with interactive dashboard processing **25,000+ warehouse entities**

Tech Stack: Python, Neo4j, GraphRAG, Groq API (LLaMA-3.3-70B), Streamlit, Pandas, Loguru, FastAPI

Multi-Store Demand Forecasting System | GitHub

Business Problem: Pharmacy chains faced **40%+ stockouts** and significant wastage due to unpredictable SKU-level demand across stores, with traditional ARIMA models failing to capture seasonality and store-item interactions.

Business Solution: Developed end-to-end deep learning platform using **CNN-LSTM architecture** with trend-seasonality features for **500+ SKU forecasts**, deployed with real-time FastAPI backend and interactive Streamlit dashboard for **30-day demand predictions**.

Achievements:

- Achieved **16.49% MAPE**, outperforming ARIMA baseline by **35%**
- Reduced **stockouts by 40%** and **wastage by 30%**, generating **\$150K annual savings**
- Accelerated decision-making by **90%** through automated forecasting workflows
- Successfully deployed scalable system handling **500+ SKU-store combinations**

Tech Stack: Python, TensorFlow/Keras, CNN-LSTM, FastAPI, Streamlit, Pandas, NumPy, Plotly, Docker

Skills

Languages: Python, SQL

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly, TensorFlow, Keras, PyTorch, Hugging Face Transformers, LangChain, LangGraph, FastAPI, Flask, Streamlit, REST APIs (CRUD)

Data Processing & Analysis: Data cleaning, preprocessing, manipulation, aggregation, validation, EDA, trend and correlation analysis, data visualization

Machine Learning & Deep Learning: Regression, Classification, Clustering, Random Forest, Gradient Boosting, SVM, CNN, RNN, LSTM, Transformers, time-series forecasting, hyperparameter tuning, model optimization

LLMs & Generative AI: LLMs, embeddings, RAG, GraphRAG, agentic AI workflows, prompt engineering, fine-tuning (LoRA, QLoRA)

Databases: MySQL, PostgreSQL, Neo4j, ChromaDB

Cloud & MLOps: Microsoft Azure (Cognitive Services, Azure OpenAI), Docker, MLflow, GitHub Actions

Messaging & APIs: Twilio, ElevenLabs, Mistral OCR, Gemini LLM, OpenAI API, Groq API, RabbitMQ, Celery

Education

Avanthi Degree and PG College, Hyderabad

B.Sc. in Mathematics, Statistics, and Computer Science

Certifications

- Data Science Certification Program – 360DigitMG
- Introduction to Programming Using Python – 360DigitMG
- Basic Course on SQL – 360DigitMG
- Analyzing and Visualizing Data with Microsoft Power BI – 360DigitMG
- Introduction to LangGraph – LangChain Academy
- The State University of New York (SUNY), Potsdam, USA